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Electronic supplementary material

Payoff scale reshapes how language models play the prisoner's dilemma

Contents

A. Supplement to the methods

A.1 Scope of the corpus and the completeness guard

The corpus is a complete factorial crossing of six language models, ten payoff scales, five prompt languages, four ordered persona pairings and ten replicates. The unit of collection is the dyad, one ten-round game between two agents of the same model; it contributes two agent-games, an agent-game being the ten decisions one agent takes in one game. Each of the 300 model-by-language-by-scale cells holds exactly 40 dyads, so the corpus is 12,000 dyads, 24,000 agent-games and 240,000 decisions (table B.1). The five models the main text reports carry 250 of those cells, 10,000 dyads, 20,000 agent-games and 200,000 decisions.

Balance is enforced rather than assumed. The analysis build refuses to write any table unless two conditions hold. First, every model-by-language-by-scale cell contains exactly 40 dyads, with none missing and none duplicated. Second, the payoffs recorded in each game recover, to a relative tolerance of 10^{-6}, the payoff scale that cell was run at, which catches a mislabelled or misfiled run rather than trusting the directory name. The guard exists because the corpus was collected in batches over several sessions and an incomplete batch is otherwise invisible in an aggregate.

The polarity of the action mapping is load-bearing for every cooperation rate in this article, so it was checked twice and in two ways. In the prompt, Option A pays less in both columns of a penalty matrix that the agent is told to minimise, and is therefore the dominant, defecting action, while mutual Option B at 2λ beats mutual Option A at 6λ and is the cooperative one. In the data, the realised penalties recorded for each pair of actions reproduce that matrix. Cooperation means Option B throughout.

The empirical logs confirm that the models experience the actual strategic tension of the prisoner's dilemma rather than an artifact of prompt confusion. Own cooperation correlates with own utility at Spearman $\rho = -0.015$, confirming that unilateral cooperation is privately unrewarding. Conversely, the opponent's cooperation correlates with own utility at Spearman $\rho = +0.916$, confirming that the partner's move governs individual payoff. At the dyad level, total welfare correlates with joint cooperation at Spearman $\rho = +0.985$, demonstrating that mutual cooperation maximizes collective surplus. The empirical incentive structure thus faithfully reproduces the theoretical dilemma.

A.2 Endpoints, seeds, parsing and fallback

The six models were served through the Kaggle Benchmarks model proxy on an OpenAI-compatible chat-completions endpoint, at temperature 1.0, with the identifiers listed in table A.1. Four of the six identifiers carry a date and two do not, which is the distinction the main text draws on in §2.3: an undated identifier can in principle be re-pointed by the provider, and one that is additionally marked as a preview is announced as provisional.

Table A.1. Endpoint identifiers. As addressed through the Kaggle Benchmarks model proxy during collection. The last column records whether the identifier fixes a build. Gemini 3.1 Flash-Lite is reported in §B.7 rather than in the main text because its identifier is both undated and marked as a preview.

Model as printed	Endpoint identifier	Dated
Claude Haiku 4.5	`claude-haiku-4-5-20251001`	yes
GPT-5.4 Nano	`gpt-5.4-nano-2026-03-17`	yes
Gemini 3.5 Flash-Lite	`gemini-3.5-flash-lite`	no
Qwen3 235B-A22B	`qwen3-235b-a22b-instruct-2507`	yes
Grok 4.20	`grok-4.20-0309-non-reasoning`	yes
Gemini 3.1 Flash-Lite	`gemini-3.1-flash-lite-preview`	no, preview

Sampling runs under a common-random-numbers scheme: a given round of a given replicate is seeded identically at every payoff scale, so a difference between two scales is not a difference between two independent samples. This reduces the variance of a within-replicate comparison across scales; it does not affect the permutation test, which shuffles the scale label among whole dyads and therefore treats the seed as part of the design.

Decisions are elicited as free text and parsed by the protocol's own matcher. A reply the matcher cannot resolve is regenerated, up to two further attempts, and if all three attempts fail the decision is recorded as Option A, which under this framing is defection. The direction of the resulting bias is therefore known: the fallback can only push a measured cooperation rate down, never up, and it cannot manufacture the cooperative behaviour reported in the main text. Each collection run checks its own fallback rate against a tolerance of 0.02 once the sweep has finished, and records a failed assertion in its benchmark report if the rate exceeds it. That check flags a run rather than halting it, and the achieved rates are not carried in the released per-game tables, so we cannot report how they vary with the payoff scale. If the fallback rate were itself to rise with the printed magnitude of the payoffs, it would bias the smallest measured ranges downward rather than create them.

A.3 The scaled payoff matrix at all ten scales

The manipulation multiplies the four cells of the penalty matrix by a positive constant λ and changes nothing else in the prompt. With base penalties $T = 0$, $R = 2$, $P = 6$, $S = 10$ the agent is shown the values in table A.2. Two features of the rendering matter for interpretation. The temptation payoff prints as 0 at every scale, so only three of the four numbers move. And the largest number shown is 10λ, so every payoff printed is at most 1 exactly when $\lambda \leq 0.1$, which two of the ten scales satisfy; that is the sub-unit boundary the persona result in §3.4 is organised around, and it is a fact about the rendering rather than a free parameter.

Table A.2. The payoff matrix as printed at each of the ten scales. Values are the penalties printed to the agent, which the prompt instructs it to minimise, in the temptation, reward, punishment and sucker notation. The horizontal rule marks the sub-unit boundary, above which some printed payoff exceeds 1.

λ	$T = 0\lambda$	$R = 2\lambda$	$P = 6\lambda$	$S = 10\lambda$
0.01	0	0.02	0.06	0.1
0.1	0	0.2	0.6	1
0.25	0	0.5	1.5	2.5
0.5	0	1	3	5
1	0	2	6	10
2	0	4	12	20
5	0	10	30	50
10	0	20	60	100
100	0	200	600	1000
1000	0	2000	6000	10000

A.4 Prompt templates and a framing artefact

The prompt templates are byte-identical to the resource files of the FAIRGAME protocol [1] and were checked against them before every collection run, by a test that compares the rendered template with the resource file byte for byte. The check exists because the templates carry Arabic, Chinese and Vietnamese text inline, and a single save in a non-UTF-8 encoding silently corrupts the prompt actually sent to the model rather than only a comment.

Artefacts of the published templates are reproduced rather than repaired, so that the corpus matches the protocol as published. The consequential one is that the Arabic and Chinese versions state the four cells as penalties while phrasing the goal sentence as maximising a reward. We therefore treat the five languages as five distinct stimuli rather than as five renderings of one, and we do not attribute a difference in the *level* of cooperation between two languages to the language alone. The wording is fixed across the ten payoff scales, so it cannot generate a dependence on the scale, and every scale-sensitivity quantity reported in the main text is computed within a language.

The persona line is a single sentence in the language of the prompt, naming the agent's own disposition and never the opponent's. In English the two forms are *You are cooperative.* and *You are selfish.* Because the opponent's persona is never disclosed, a persona pairing reaches a player only through realised behaviour, and not before round 2.

A.5 The rule base

Each agent-game is encoded as a sequence of ten two-letter tokens of the form ⟨previous outcome⟩⟨current action⟩, over the outcome letters E (no history yet), R (both cooperated), S (the player cooperated and was exploited), T (the player exploited a cooperator) and P (both defected), and the actions C and D. The vocabulary is therefore eleven symbols including padding, and it is shared exactly between the synthetic training corpus and the language-model transcripts, which is what makes a classifier trained on the former applicable to the latter.

All four canonical strategies are memory-one, so each prescribes an action as a function of the previous outcome letter alone (table A.3). Deciding whether a trajectory is *exactly* what a strategy would have played on the history that actually occurred is then a per-token lookup, with no learning and no tolerance parameter.

Table A.3. The four canonical memory-one rules. The action each rule prescribes given the previous round's joint outcome. Tit-for-Tat and Win-Stay-Lose-Shift differ only in the two columns T and P, which is why ten rounds of mutual cooperation are consistent with AllC, Tit-for-Tat and Win-Stay-Lose-Shift at once.

	E (first move)	R	S	T	P
AllC	C	C	C	C	C
AllD	D	D	D	D	D
Tit-for-Tat	C	C	D	C	D
Win-Stay-Lose-Shift	C	C	D	D	C

Formally, writing $\mathbf{a} = (a_1, \dots, a_N)$ for the agent's actions and $\mathbf{o} = (o_1, \dots, o_N)$ for the opponent's, and $\pi_k(\cdot)$ for the action prescribed by rule k in table A.3, the match is

$$\text{consistent}(\mathbf{a}, \mathbf{o}, k) \equiv \bigwedge_{t=1}^{N} \left[a_t = \pi_k(e_t) \right], \qquad e_1 = E, \quad e_t = \text{outcome}(a_{t-1}, o_{t-1}) \ \ (t \geq 2),$$

and the deviation count that the main text calls the rule distance is the same expression with the conjunction replaced by a sum of indicators, $d_k = \sum_{t=1}^{N} \mathbb{I}[a_t \neq \pi_k(e_t)]$, reported as $\min_k d_k$, the number of rounds violating the nearest canonical rule.

Because several rules routinely prescribe the same actions on the history that occurred, the matcher returns a *set* rather than a label. A player who cooperated in every round against a cooperator is exactly consistent with AllC, Tit-for-Tat and Win-Stay-Lose-Shift simultaneously, and no method, learned or otherwise, can separate them from that evidence; reporting the set is the correct answer rather than a failure of the method. The three provenances are recorded separately and never merged:

- *deduced*, where exactly one rule fits, so the label is a deduction;
- *ambiguous*, where several fit, so the rules fix a candidate set and the classifier only ranks inside it;
- *unmatched*, where no rule fits, so the label is an attribution the classifier makes alone.

There is no confidence threshold anywhere in this read-out, and no composite label is emitted: the multi-label answer is the set itself, and where a single label is needed for a table it is the classifier's ranking restricted to that set. The classifier can therefore never overturn a deduction.

A.6 Classifier training and validation

The learned branch is trained on a synthetic corpus that contains no language-model output at all. Trajectories of the four canonical strategies are generated against random opponents at four execution-noise levels, balanced at 40,320 sequences per strategy per level, giving 161,280 sequences at each level. Training uses the zero-noise and 5% levels pooled, 322,560 sequences split 80/20 at random, so the held-out split has 64,512 sequences; the 10% and 20% levels are never trained on and serve as out-of-distribution tests.

The network is a single-layer unidirectional LSTM [2]: an embedding of the eleven-symbol vocabulary into 24 dimensions, a recurrent layer of 96 hidden units, dropout 0.15 and a linear map to the four classes. It is optimised with AdamW at learning rate 3×10^{-3} and weight decay 10^{-4} under a cosine schedule, for 12 epochs at batch size 512, with seed 1234. Accuracy is 0.985 on the held-out split, 0.936 at 10% execution noise and 0.811 at 20%, with per-class recall on the held-out split of 0.985 for AllC, 0.995 for AllD, 0.974 for Tit-for-Tat and 0.983 for Win-Stay-Lose-Shift (figure A.1 and table B.10). The two conditional rules are the harder pair, as expected: an execution error in either can be read as the signature of the other.

The 20% level bounds where the read-out is warranted, and §D.3 states where the corpus sits relative to it.

A.7 Resampling and testing

Three dependencies rule out treating agent-games as independent: the two agents of a dyad play each other, the ten rounds inside an agent-game are strongly autocorrelated, and the ten replicates of a cell share a payoff scale. Every interval reported in this article is therefore a nonparametric bootstrap that resamples *whole dyads* with replacement, 1,500 draws, percentile method [3], with a fixed seed so that draws are common-random-number paired across cells. Every regression clusters its standard errors on the dyad.

The range of a model's cooperation over the ten scales is a maximum minus a minimum and so is positive by construction. Its bootstrap interval quantifies the precision of the range, not its distance from zero, and the test is supplied

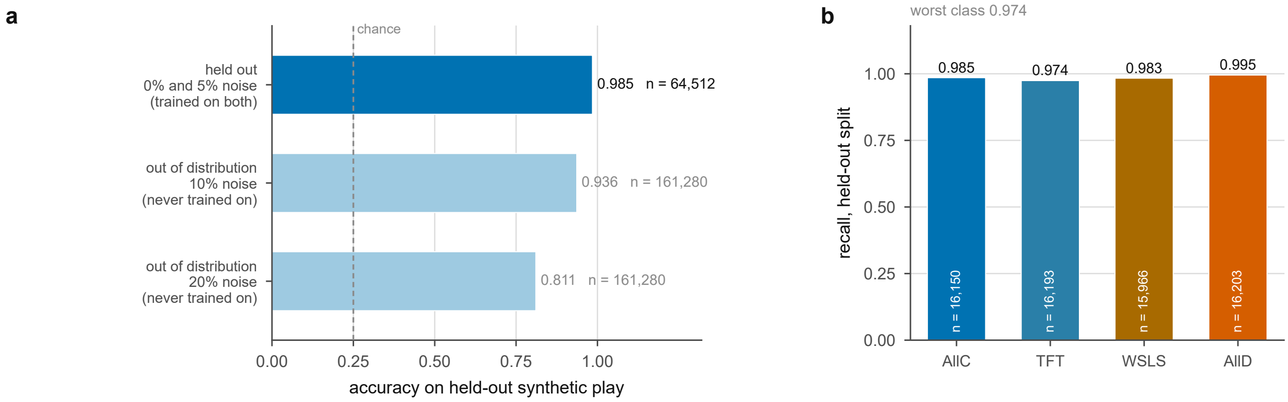

Figure A.1. How well the learned branch recovers a rule that is known by construction. (a) Accuracy on the held-out split of the training noise levels and on the two noise levels never trained on, against the chance level of a balanced four-way problem. (b) Per-class recall on the held-out split. The two conditional rules are the harder pair: an execution error in either can be read as the signature of the other.

separately by a permutation. Because the null is a theorem rather than a nuisance hypothesis, the permutation destroys exactly the association under test and leaves the rest of the design as observed: the payoff-scale label is shuffled among whole dyads within language-by-pairing strata, the unit to which the scale was assigned, and the range of the ten scale means is recomputed. With 2,000 draws the smallest attainable p value is $1/2001 = 0.0005$. Shuffling agent-games independently would be anti-conservative, because the two agent-games of a dyad would then be free to land in different scale groups although they were generated together.

The variance decomposition is a type-II analysis of variance on cell means with terms for model, language and payoff scale and their three two-way interactions, and no three-way term, so the residual is the omitted three-way interaction: 144 degrees of freedom on the 250 cells of the five main-text models, and 180 on all 300 cells.

B. Supplement to the results

B.1 Cooperation by model and payoff scale

Table B.2 gives the cooperation rate of every model at every payoff scale, with a bootstrap interval under each entry, for all six models. It is the numerical form of figure 2b.

B.2 Cooperation by model, prompt language and payoff scale

Table B.3 gives the same quantity broken down by prompt language, 40 dyads per entry, with the across-scale range in the final column. Over all six models that range runs from 0.080, for Gemini 3.1 Flash-Lite in Arabic, to 0.855, for Grok 4.20 in English. It is the numerical form of figure 3.

B.3 Variance decomposition

Table B.4 is the type-II analysis of variance on the 250 cell means of the five main-text models, and table B.5 the same decomposition on all 300 cells. Every term significant in the first is significant in the second and in the same order, and the interaction of the payoff scale with the model again carries several times the variance the scale carries as a main effect.

A third fit bears on how the main effect should be read. Refitting the five-model decomposition with Grok 4.20 removed leaves the payoff-scale main effect not significant, at $F_{9,108} = 1.15$, $p = 0.33$ and 0.9% of the variance across the remaining 200 cells, while every one of those four models still departs from the invariance on its own (table 1). The pooled main effect is therefore carried by a single member of the panel: it is a statement about that model surviving the averaging rather than a property of language models in general, which is why the main text reports it as such rather than as the headline.

B.4 Persona, in full

Table B.6 gives the persona effect, the cooperation rate under a cooperative instruction minus that under a selfish one, at each payoff scale and for all six models. The two leftmost columns are the scales at which every printed payoff is at most 1.

Two models invert the instruction at every scale rather than only below the sub-unit boundary. Pooled over the grid, Qwen3 235B-A22B cooperates 0.072 under the cooperative persona and 0.798 under the selfish one, and Grok 4.20 inverts less extremely but throughout. The most likely reading is that the story the prompt tells is ambiguous about whom one is cooperating with, one's partner in crime or the interrogating authority, and that these two models resolve that ambiguity opposite to the convention the payoff matrix encodes. The inversion is not the subject of this article and we make no claim about its cause; we report it because an analysis that quietly excluded the two models whose persona response runs backwards would flatter the persona result. Its magnitude, like everything else measured here, depends on the payoff scale, and the per-scale sequence is not monotone, so no trend is asserted.

B.5 The strategy read-out, in full

Table B.7 gives the share of agent-games matched exactly by one canonical rule, by model and payoff scale, computed by exact matching with no learned component. Table B.8 gives the four-way label composition over the same grid.

Table B.9 is the one that bounds how the composition may be read. For each label it gives the share of the agent-games carrying it that were deduced, ambiguous or unmatched. The asymmetry is severe: AllC and AllD carry a substantial deduced share, at 23.6% and 35.6% over the corpus, whereas 94.6% of the Tit-for-Tat labels and 97.1% of the Win-Stay-Lose-Shift labels are attributions made where no canonical rule fits at all. A movement in the Tit-for-Tat share is therefore a statement about which canonical rule the play most resembles, and not evidence that any model is running Tit-for-Tat. The two claims the main text makes about rule agreement, the falling deduced share and the rising rule distance, do not depend on the classifier at all.

B.6 Why the model curves cancel under pooling

Tables B.11 and B.12 give the pairwise correlation between the ten-point cooperation-against-scale curves of each pair of models, for the five main-text models and for all six.

The cancellation is arithmetic. Writing $c_m(\lambda)$ for model m's curve and $\bar{c}(\lambda)$ for their average, the range of $\bar{c}$ is bounded above by the average of the individual ranges, with equality only if the curves attain their extremes at the same scales and move together in between. Here six of the ten pairs are anticorrelated and the mean correlation is -0.10, so the bound is far from tight: the pooled range is 0.154 against a mean within-model range of 0.249, an attenuation of 1.62. With the sixth model included the same quantities are 0.125, 0.219 and 1.76, with nine of the fifteen pairs negative, so the pooling argument is stronger over the whole corpus than over the five models the main text reports.

This is not a power problem. Collecting more dyads tightens the interval around $\bar{c}$; it does not make $\bar{c}$ a better description of any c_m.

The cancellation of model curves under pooling reflects an underlying structural divide in how models play: per-game cooperation is not a graded Gaussian quantity, but a mixture over distinct policy regimes (figure B.1). Across all 24,000 agent-games, 11.1% never cooperate and 28.5% always cooperate, so that two discrete boundary atoms carry 39.5% of all play against 18.2% expected under a flat distribution across the 11 attainable rates.

Crucially, this boundary concentration is a model-level trait rather than an invariant property of the task. In Claude Haiku 4.5, boundary mass accounts for only 8.6% of play; its distribution is unimodal with an interior mode at 0.2, reflecting an adaptive policy that explores intermediate action frequencies. In stark contrast, Qwen3 235B-A22B (60.4% boundary mass) and Grok 4.20 (59.8% boundary mass) play predominantly degenerate pure strategies, switching between always cooperating and always defecting. A single pooled mean cooperation rate therefore conflates two distinct phenomena: how often a model commits to an unconditional pure policy, and which pure policy it selects.

B.7 The sixth model, Gemini 3.1 Flash-Lite

The sixth model was run on the identical grid, with the identical protocol, and every quantity the main text reports for the five is reported for it here and in the tables of this section, which cover all six models throughout.

Why it is reported here rather than in the main text. Its endpoint identifier, `gemini-3.1-flash-lite-preview`, is both undated and marked by the provider as provisional, so what was served under that name while the corpus was collected cannot be re-addressed afterwards and a reader cannot reproduce the run against the same object. We state the cost of drawing the line there rather than leaving it to be found. The identifier of Gemini 3.5 Flash-Lite, which stays in the main text, is undated as well and could in principle be re-pointed to a later build, so it carries a weaker form of the same risk; the line drawn here is between an identifier marked provisional and one that is not, and not between pinned and unpinned. The decision was taken on the identifier and not on the results.

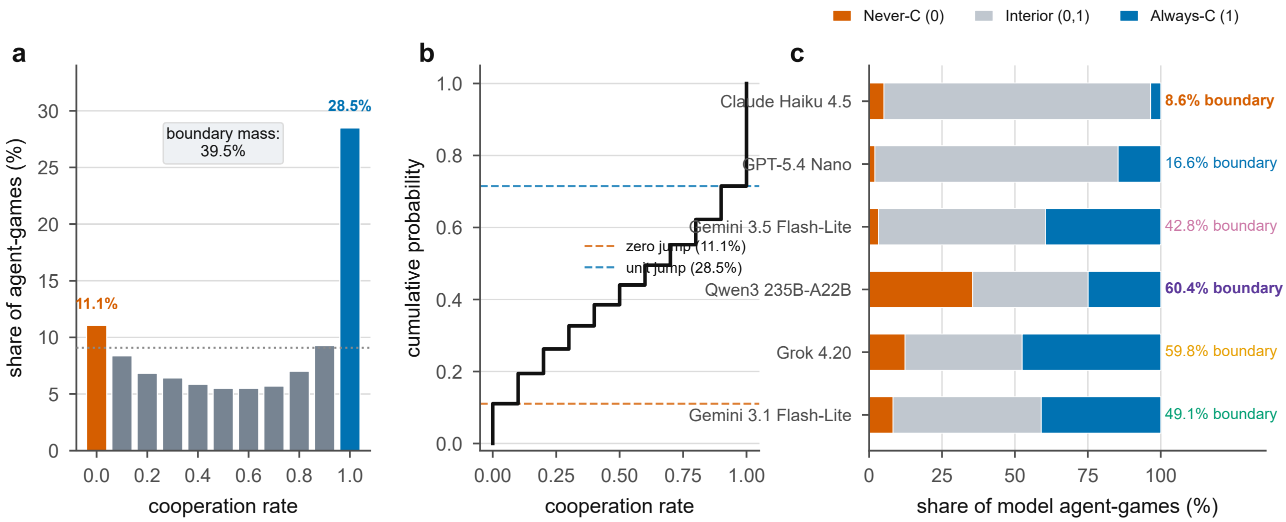

Figure B.1. Bimodal cooperation distribution and model-level policy heterogeneity. (a) Pooled cooperation distribution across 24,000 agent-games, showing extreme boundary inflation: 11.1% of games never cooperate and 28.5% always cooperate, totaling 39.5% boundary mass compared to 18.2% under a flat reference benchmark. (b) Empirical cumulative distribution function (ECDF), displaying the steep vertical steps at the defection and cooperation boundaries. (c) Decomposition into never-cooperate ($c = 0$), interior mixed play ($0 < c < 1$), and always-cooperate ($c = 1$) across all six models. The boundary share varies seven-fold across models, from 8.6% in Claude Haiku 4.5 (unimodal interior distribution) to 60.4% in Qwen3 235B-A22B and 59.8% in Grok 4.20 (predominantly degenerate pure-policy play).

What it does. Its cooperation range across the ten scales is 0.072 (95% CI 0.060, 0.133), the smallest in the corpus, with a permutation $p = 0.0005$ and a clustered Wald statistic of 83.1 on nine degrees of freedom, so it departs from the invariance on the same evidence as the other five. Its across-scale range by language runs from 0.080 in Arabic to 0.170 in Vietnamese. Its persona effect is $+0.577$ below the sub-unit boundary and $+0.680$ above it, a shift of 0.103 (95% CI 0.075, 0.131) with no sign reversal. Its opening-move range is 3.68 times its range over rounds 2 to 10, the largest ratio in the corpus, so the localisation of the effect at the first move is more pronounced in it than in any main-text model, not less.

The one result it reverses. It is the only model of the six whose share of agent-games matched exactly by one canonical rule *rises* with the payoff scale, from 17.8% below the sub-unit boundary to 26.3% above it, with a significant positive trend of $\beta = +0.125$ per decade of λ; its rule distance is flat ($\beta = -0.019$, $p = 0.26$); figure B.2 draws it beside the five. The main text's statement that play becomes less rule-governed in four models of five therefore reads, over the whole corpus, as four models of six moving one way, one showing no trend and one moving the other. That is the form the evidence supports, and the main text says so at the point where the claim is made.

B.8 The opening move, in detail

Table B.13 gives, for all six models, the across-scale range of cooperation on the opening move and over rounds 2 to 10, and their ratio. The two windows rest on 400 and 3,600 decisions per cell respectively, so their ranges are not variance-matched: a range computed from fewer observations is upward-biased, and the ratios are therefore descriptive rather than tests. The comparison is still informative because the bias runs the same way in every model while the ratio does not, spanning 0.47 in Qwen3 235B-A22B to 3.68 in Gemini 3.1 Flash-Lite.

Figure B.3 identifies the mechanism through which the payoff multiplier alters behaviour. The multiplier acts by re-weighting the probabilities of unconditional policies before play begins, rather than by reshaping within-game backward induction or adaptive learning. Across the ten-scale ladder, the share of agents committing to unconditional defection falls from 21.3% at $\lambda = 0.01$ to 8.9% at $\lambda = 1000$ (Spearman $\rho = -0.47$), with the sharpest transition occurring across the sub-unit boundary $\lambda \leq 0.1$. Meanwhile, the unconditional cooperation share rises modestly from 24.2% to 29.0% (Spearman $\rho = +0.05$).

In contrast, the within-game trajectories across rounds 1 to 10 remain near-parallel across five orders of magnitude: games run at $\lambda = 0.01$, $\lambda = 1.0$, and $\lambda = 1000$ exhibit virtually identical dynamic decay profiles (figure B.3c). The payoff scale therefore acts as an ex-ante stance selector on the opening move, establishing the cooperative climate of the dyad before any interaction history has elapsed.

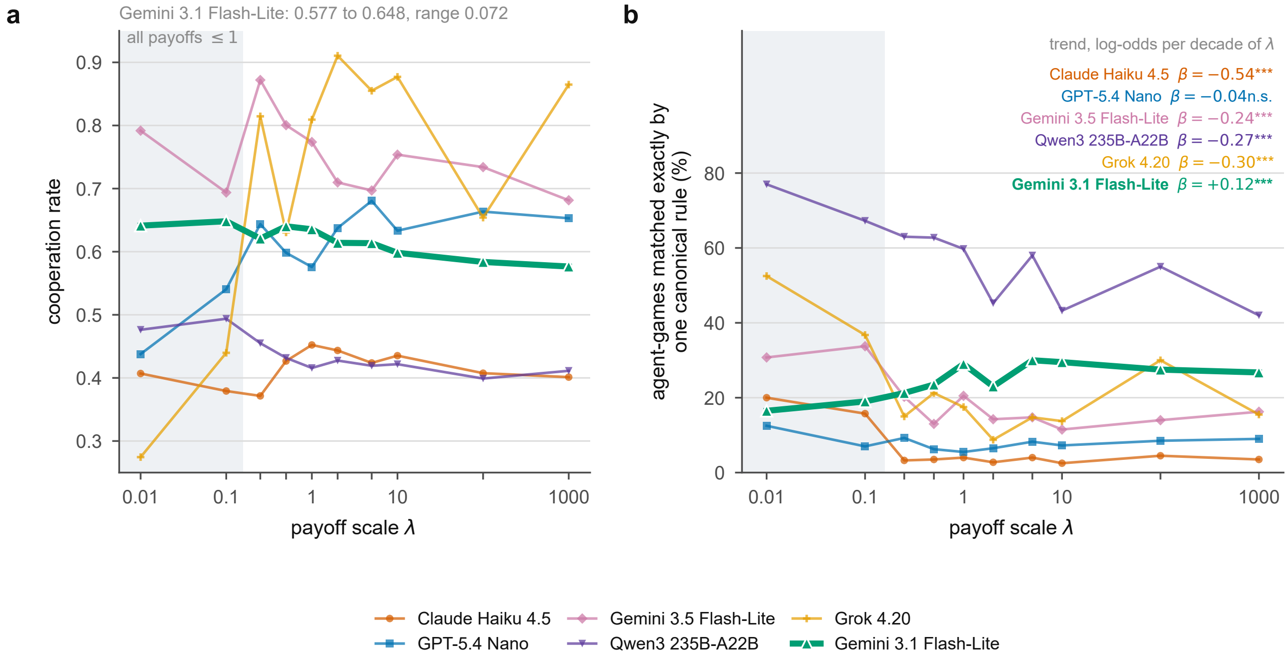

Figure B.2. The sixth model beside the five reported in the main text. Gemini 3.1 Flash-Lite is drawn heavier and in front. (a) Cooperation against the payoff scale, where it sits inside the spread of the other five. (b) Share of agent-games matched exactly by one canonical rule, with each model's trend per decade of λ. It is the only model of the six whose trend is positive.

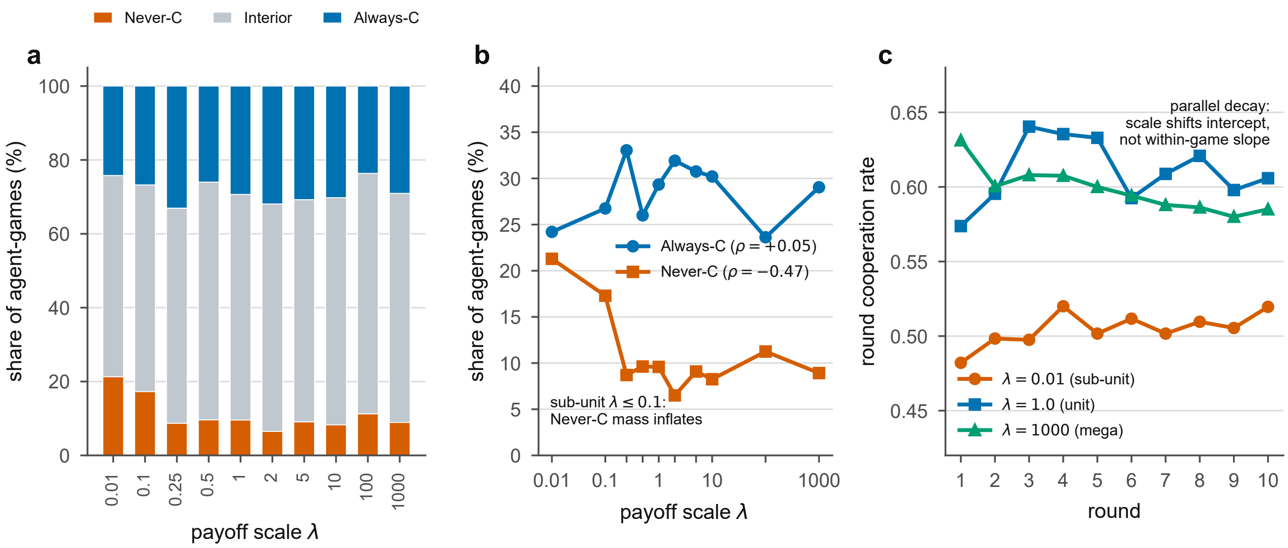

Figure B.3. The scaling mechanism: ex-ante policy re-weighting versus within-game dynamics. (a) Share of agent-games playing never-cooperate ($c = 0$), interior mixed strategies, and always-cooperate ($c = 1$) across the ten payoff scales λ. The never-cooperate share drops from 21.3% at $\lambda = 0.01$ to 8.9% at $\lambda = 1000$. (b) Opposing trajectories of the two boundary atoms against $\log_{10} \lambda$: the always-cooperate share rises modestly (Spearman $\rho = +0.05$), while the never-cooperate share falls substantially (Spearman $\rho = -0.47$), showing that the scale effect is concentrated in the relief of unconditional defection at sub-unit stakes. (c) Round-by-round cooperation rate across rounds 1 to 10 for three anchor scales ($\lambda = 0.01$, $\lambda = 1.0$, and $\lambda = 1000$). The three decay curves remain near-parallel throughout the game, establishing that payoff magnitude sets the initial stance at round 1 rather than modifying within-game dynamic adaptation.

B.9 Outcome fairness, net exploitation and the V-shaped welfare frontier

Studies evaluating machine behaviour in multi-agent environments often report outcome fairness, measured via the Gini coefficient or payoff disparity between paired agents, interpreting lower disparity as evidence of egalitarian or prosocial alignment. In symmetric normal-form games played between identical agents, however, within-dyad inequality is not an independent behavioural dimension: it is an exact algebraic function of one-directional exploitation.

Because both agents face identical payoff matrices, whenever both choose the same action (mutual defection Option A or mutual cooperation Option B), their per-round payoffs are identical by construction ($\pi_1 = \pi_2$). A payoff disparity can only arise in asymmetric rounds where one agent cooperates while the other defects. Across a ten-round repeated interaction, the relationship is exact:

$$\text{signed\_gap} = p_{DC} - p_{CD}, \qquad \text{utility\_gap} = |p_{DC} - p_{CD}| = p_{\text{exploit}} - 2\min(p_{CD}, p_{DC}), \tag{A 1}$$

where p_{DC} and p_{CD} denote the proportion of rounds in which the focal agent exploits or is exploited by its partner, respectively. Across all 12,000 dyads in the corpus, the maximum residual from this identity is zero. Over 72.8% of dyads end with zero payoff gap, and the observed gap correlates with total one-directional exploitation at $r = 0.941$. Within-dyad inequality therefore measures coordination failure rather than a distributional preference.

Figure B.4 illustrates the resulting relationship between fairness and joint welfare. Rather than a monotonic trade-off between equality and efficiency, the empirical frontier traces a distinct V-shaped curve. In the poorest welfare tercile, dyads are locked in mutual defection (DD), yielding near-perfect equality (fairness $= 1 - \text{Gini} \approx 0.98$) at minimal social surplus. As dyads begin to explore cooperation without mutual coordination, unilateral exploitation spikes, causing fairness to reach its minimum (0.62). Finally, as dyads achieve mutual cooperation (CC), joint welfare peaks alongside a return to perfect equality (fairness ≈ 1.0).

Crucially, this structural coordination mechanism explains why the payoff scale exerts zero influence on within-dyad fairness: the regression slopes of both the within-dyad utility gap and Gini coefficient against $\log_{10}\lambda$ are flat at 0.0004 ($p = 0.80$). Payoff scaling shifts the aggregate willingness to cooperate, but leaves the symmetric coordination balance between paired clones entirely untouched.

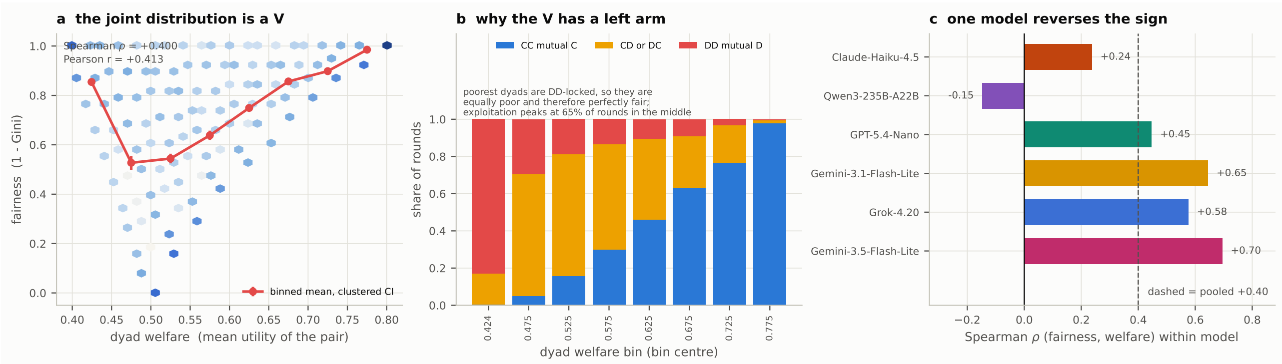

Figure B.4. Outcome fairness, net exploitation and the V-shaped welfare frontier. (a) Within-dyad fairness ($1 - \text{Gini}$) plotted against joint dyad welfare across all experimental conditions, revealing a characteristic V-shaped relationship where equality peaks at both mutual defection (low welfare) and mutual cooperation (high welfare), with inequality concentrated in the exploitative transition zone. (b) Per-model fairness trajectories showing how different models occupy distinct regimes along the welfare-fairness frontier. (c) Invariance of distributional equity across the ten payoff scales: within-dyad inequality remains constant across five orders of magnitude (slope 0.0004, $p = 0.80$), confirming that scale alters overall cooperation rates without destabilising dyadic symmetry.

Table B.1. The corpus, by model. Every model was run on the same ten payoff scales and five prompt languages. "Cell n" is the number of dyads in each model × language × scale cell; the design is balanced with none missing, and the build refuses to write its tables otherwise. The first five models are the ones the main text reports; Gemini 3.1 Flash-Lite is reported here, for the reason given with the sixth model below.

Model	Scales	Lang.	Dyads	Agent-games	Decisions	Cell n	Coop.
Claude Haiku 4.5	10	5	2,000	4,000	40,000	40	0.415
GPT-5.4 Nano	10	5	2,000	4,000	40,000	40	0.606
Gemini 3.5 Flash-Lite	10	5	2,000	4,000	40,000	40	0.751
Qwen3 235B-A22B	10	5	2,000	4,000	40,000	40	0.435
Grok 4.20	10	5	2,000	4,000	40,000	40	0.713
Gemini 3.1 Flash-Lite	10	5	2,000	4,000	40,000	40	0.617
All	10	5	12,000	24,000	240,000	40	0.590

Table B.2. Cooperation by model and payoff scale. Each entry aggregates 200 dyads. The bracketed line under each model is a 95% confidence interval from a bootstrap that resamples whole dyads.

Model	λ=0.01	λ=0.1	λ=0.25	λ=0.5	λ=1	λ=2	λ=5	λ=10	λ=100	λ=1000
Claude Haiku 4.5	0.407	0.379	0.371	0.427	0.452	0.444	0.424	0.435	0.407	0.401
	[0.37,0.45]	[0.34,0.42]	[0.34,0.40]	[0.39,0.46]	[0.42,0.49]	[0.41,0.48]	[0.39,0.46]	[0.40,0.47]	[0.38,0.43]	[0.37,0.43]
GPT-5.4 Nano	0.438	0.540	0.644	0.598	0.576	0.637	0.680	0.633	0.663	0.653
	[0.40,0.48]	[0.50,0.58]	[0.61,0.68]	[0.56,0.63]	[0.54,0.61]	[0.60,0.67]	[0.65,0.71]	[0.60,0.67]	[0.64,0.69]	[0.62,0.68]
Gemini 3.5 Flash-Lite	0.792	0.694	0.872	0.800	0.774	0.710	0.697	0.754	0.734	0.682
	[0.74,0.84]	[0.64,0.74]	[0.85,0.89]	[0.77,0.83]	[0.75,0.80]	[0.67,0.74]	[0.66,0.73]	[0.72,0.78]	[0.70,0.77]	[0.64,0.72]
Qwen3 235B-A22B	0.476	0.494	0.455	0.432	0.416	0.427	0.419	0.422	0.399	0.411
	[0.43,0.52]	[0.44,0.54]	[0.41,0.50]	[0.39,0.47]	[0.38,0.45]	[0.39,0.46]	[0.38,0.45]	[0.39,0.45]	[0.37,0.43]	[0.38,0.44]
Grok 4.20	0.274	0.440	0.815	0.630	0.809	0.910	0.855	0.877	0.654	0.865
	[0.23,0.32]	[0.39,0.49]	[0.77,0.85]	[0.59,0.67]	[0.77,0.85]	[0.88,0.94]	[0.82,0.89]	[0.84,0.91]	[0.61,0.70]	[0.83,0.90]
Gemini 3.1 Flash-Lite	0.641	0.648	0.621	0.640	0.635	0.614	0.614	0.598	0.584	0.577
	[0.61,0.67]	[0.62,0.68]	[0.59,0.65]	[0.61,0.67]	[0.61,0.67]	[0.58,0.65]	[0.58,0.65]	[0.56,0.63]	[0.54,0.62]	[0.54,0.61]

Table B.3. Cooperation by model, prompt language and payoff scale. 40 dyads per entry. The final column is the range across the ten scales for that model and language; over all six models it runs from 0.080, for Gemini 3.1 Flash-Lite in Arabic, to 0.855, for Grok 4.20 in English, so the scale sensitivity of a model is a property of the model and the language jointly.

Model	Language	0.01	0.1	0.25	0.5	1	2	5	10	100	1000	Range
Claude Haiku 4.5	English	0.35	0.40	0.32	0.42	0.44	0.45	0.41	0.42	0.45	0.46	**0.136**
	French	0.65	0.47	0.50	0.33	0.37	0.43	0.41	0.40	0.38	0.40	**0.314**
	Vietnamese	0.17	0.17	0.28	0.29	0.26	0.20	0.30	0.22	0.23	0.23	**0.130**
	Chinese	0.53	0.52	0.45	0.61	0.69	0.57	0.56	0.66	0.58	0.56	**0.245**
	Arabic	0.35	0.33	0.31	0.48	0.51	0.56	0.43	0.47	0.40	0.36	**0.249**
GPT-5.4 Nano	English	0.45	0.64	0.85	0.88	0.86	0.90	0.88	0.87	0.89	0.76	**0.449**
	French	0.75	0.78	0.78	0.57	0.55	0.60	0.62	0.67	0.55	0.74	**0.226**
	Vietnamese	0.49	0.46	0.58	0.47	0.50	0.62	0.75	0.54	0.65	0.63	**0.289**
	Chinese	0.24	0.45	0.65	0.60	0.49	0.57	0.61	0.58	0.62	0.65	**0.414**
	Arabic	0.25	0.37	0.37	0.48	0.47	0.49	0.54	0.51	0.61	0.49	**0.360**
Gemini 3.5 Flash-Lite	English	0.94	0.66	0.92	0.84	0.74	0.59	0.53	0.65	0.64	0.60	**0.412**
	French	0.91	0.63	0.86	0.72	0.78	0.63	0.75	0.78	0.59	0.61	**0.317**
	Vietnamese	0.72	0.63	0.93	0.86	0.81	0.76	0.62	0.64	0.89	0.73	**0.314**
	Chinese	0.69	0.80	0.76	0.64	0.70	0.73	0.67	0.76	0.66	0.73	**0.162**
	Arabic	0.70	0.75	0.89	0.94	0.83	0.84	0.92	0.93	0.89	0.74	**0.242**
Qwen3 235B-A22B	English	0.47	0.51	0.53	0.44	0.42	0.49	0.43	0.50	0.37	0.46	**0.155**
	French	0.41	0.40	0.35	0.39	0.37	0.36	0.35	0.35	0.34	0.33	**0.083**
	Vietnamese	0.52	0.55	0.48	0.48	0.45	0.45	0.47	0.44	0.40	0.42	**0.150**
	Chinese	0.48	0.52	0.46	0.44	0.43	0.45	0.45	0.44	0.49	0.47	**0.084**
	Arabic	0.50	0.50	0.46	0.42	0.40	0.40	0.40	0.38	0.39	0.38	**0.124**
Grok 4.20	English	0.14	0.24	0.76	0.71	0.88	0.97	0.93	1.00	0.72	1.00	**0.855**
	French	0.46	0.53	0.99	0.74	0.80	0.92	0.89	1.00	0.62	1.00	**0.536**
	Vietnamese	0.26	0.74	1.00	0.72	0.99	0.99	1.00	1.00	0.72	0.98	**0.740**
	Chinese	0.39	0.43	0.69	0.43	0.54	0.77	0.77	0.69	0.65	0.66	**0.381**
	Arabic	0.12	0.26	0.62	0.55	0.84	0.90	0.70	0.70	0.55	0.69	**0.786**
Gemini 3.1 Flash-Lite	English	0.74	0.62	0.67	0.62	0.63	0.62	0.59	0.62	0.57	0.62	**0.167**
	French	0.57	0.66	0.70	0.63	0.65	0.60	0.60	0.61	0.60	0.56	**0.145**
	Vietnamese	0.58	0.56	0.54	0.64	0.57	0.55	0.57	0.49	0.47	0.47	**0.170**
	Chinese	0.62	0.66	0.53	0.61	0.59	0.60	0.60	0.58	0.58	0.58	**0.133**
	Arabic	0.69	0.74	0.66	0.70	0.74	0.69	0.71	0.69	0.69	0.66	**0.080**

Table B.4. Variance decomposition of the cell means, five models. Type-II analysis of variance on the 250 model × language × scale cell means of the five models the main text reports, where the design is exactly balanced. The payoff scale carries a significant main effect, and its interaction with the model carries three times as much of the variance.

Term	Sum sq.	df	F	p	% variance
Model	4.7871	4	187.27	< 0.001	45.4
Language	0.1599	4	6.25	< 0.001	1.5
Payoff scale	0.6119	9	10.64	< 0.001	5.8
Model × language	1.7901	16	17.51	< 0.001	17.0
Model × scale	1.8910	36	8.22	< 0.001	17.9
Language × scale	0.3844	36	1.67	0.018	3.6
Residual	0.9202	144			8.7

Table B.5. Variance decomposition of the cell means, all six models. The same type-II analysis on all 300 cell means, the sixth model included. Every term significant in the five-model table is significant here and in the same order: the model × scale interaction again carries several times the variance of the payoff scale as a main effect.

Term	Sum sq.	df	F	p	% variance
Model	4.8330	5	170.26	< 0.001	44.8
Language	0.1285	4	5.66	< 0.001	1.2
Payoff scale	0.4661	9	9.12	< 0.001	4.3
Model × language	1.9467	20	17.15	< 0.001	18.0
Model × scale	2.0646	45	8.08	< 0.001	19.1
Language × scale	0.3331	36	1.63	0.020	3.1
Residual	1.0219	180			9.5

Table B.6. Persona effect by model and payoff scale. The cooperation rate of an agent instructed that it is cooperative minus that of an agent instructed that it is selfish, at each scale. The two leftmost columns are the scales at which every payoff printed is at most 1.

Model	λ=0.01	λ=0.1	λ=0.25	λ=0.5	λ=1	λ=2	λ=5	λ=10	λ=100	λ=1000
Claude Haiku 4.5	−0.211	−0.232	+0.206	+0.200	+0.179	+0.261	+0.277	+0.158	+0.168	+0.211
GPT-5.4 Nano	−0.172	−0.098	+0.015	−0.026	−0.053	+0.032	+0.041	+0.055	−0.017	+0.100
Gemini 3.5 Flash-Lite	−0.335	−0.578	+0.122	+0.234	+0.241	+0.258	+0.313	+0.203	+0.193	+0.230
Qwen3 235B-A22B	−0.901	−0.922	−0.811	−0.768	−0.702	−0.635	−0.690	−0.602	−0.657	−0.566
Grok 4.20	−0.454	−0.428	−0.263	−0.484	−0.323	−0.167	−0.235	−0.206	−0.515	−0.234
Gemini 3.1 Flash-Lite	+0.552	+0.602	+0.640	+0.670	+0.693	+0.681	+0.707	+0.696	+0.659	+0.692

Table B.7. Share of agent-games matched exactly by one canonical rule (%). Computed by exact rule matching, with no learned component. 400 agent-games per entry.

Model	λ=0.01	λ=0.1	λ=0.25	λ=0.5	λ=1	λ=2	λ=5	λ=10	λ=100	λ=1000
Claude Haiku 4.5	20.0	15.8	3.2	3.5	4.0	2.8	4.0	2.5	4.5	3.5
GPT-5.4 Nano	12.5	7.0	9.2	6.2	5.5	6.5	8.2	7.2	8.5	9.0
Gemini 3.5 Flash-Lite	30.8	33.8	20.2	13.0	20.5	14.2	14.8	11.5	14.0	16.2
Qwen3 235B-A22B	77.0	67.2	63.0	62.7	59.8	45.2	58.0	43.2	55.0	42.0
Grok 4.20	52.5	36.8	15.0	21.2	17.5	8.8	14.8	13.8	30.0	15.5
Gemini 3.1 Flash-Lite	16.5	19.0	21.2	23.5	29.0	23.0	30.0	29.5	27.5	26.8

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Table B.8. Strategy composition by model and payoff scale (%). The label assigned by the hybrid read-out: the rule base fixes the candidate set and the LSTM ranks within it, deciding alone only where no canonical rule fits. Columns sum to 100 within each model and scale.

Model	Label	0.01	0.1	0.25	0.5	1	2	5	10	100	1000
Claude Haiku 4.5	AllC	27.0	20.0	14.2	20.0	21.8	18.2	15.2	18.8	12.8	14.0
	TFT	20.5	23.5	23.5	18.0	16.8	16.5	19.5	17.8	20.8	22.2
	WSLS	8.8	13.5	15.2	13.5	20.2	21.2	20.5	18.2	21.8	19.0
	AllD	43.8	43.0	47.0	48.5	41.2	44.0	44.8	45.2	44.8	44.8
GPT-5.4 Nano	AllC	29.8	41.8	52.2	46.2	48.5	54.2	58.5	52.2	53.2	50.7
	TFT	11.8	11.5	10.5	11.8	10.2	13.2	13.8	13.2	13.0	17.8
	WSLS	12.5	12.5	14.5	15.5	13.8	10.5	12.5	10.2	15.5	14.0
	AllD	46.0	34.2	22.8	26.5	27.5	22.0	15.2	24.2	18.2	17.5
Gemini 3.5 Flash-Lite	AllC	73.8	59.8	74.0	64.0	58.0	60.0	51.7	59.8	56.2	51.5
	TFT	6.8	2.2	10.2	16.2	17.0	16.2	15.2	13.5	15.2	14.2
	WSLS	6.2	13.2	14.0	16.2	21.2	12.0	18.0	18.5	16.5	15.2
	AllD	13.2	24.8	1.8	3.5	3.8	11.8	15.0	8.2	12.0	19.0
Qwen3 235B-A22B	AllC	42.0	47.8	39.2	38.0	36.8	36.5	35.5	35.2	34.5	33.8
	TFT	0.5	0.0	0.0	0.0	1.2	0.8	2.0	6.2	0.0	4.8
	WSLS	9.0	2.5	8.8	13.5	7.2	12.0	8.5	8.2	9.2	10.0
	AllD	48.5	49.8	52.0	48.5	54.8	50.7	54.0	50.2	56.2	51.5
Grok 4.20	AllC	18.0	29.8	72.8	48.8	71.0	81.5	74.8	82.0	49.5	79.0
	TFT	5.8	10.0	9.5	9.0	4.2	5.8	6.8	3.2	8.2	4.2
	WSLS	9.8	14.8	8.5	17.5	13.5	9.0	11.5	7.0	20.0	8.0
	AllD	66.5	45.5	9.2	24.8	11.2	3.8	7.0	7.8	22.2	8.8
Gemini 3.1 Flash-Lite	AllC	55.8	57.0	58.2	61.0	59.8	55.0	58.5	53.8	52.2	52.2
	TFT	6.5	3.8	2.2	0.2	2.5	3.5	1.2	2.2	6.2	5.8
	WSLS	12.8	13.2	8.8	2.8	8.0	7.2	6.5	7.2	7.2	6.0
	AllD	25.0	26.0	30.8	36.0	29.8	34.2	33.8	36.8	34.2	36.0

Table B.9. Provenance of each label. For every label, the share of the agent-games carrying it that were deduced (exactly one canonical rule fits), ambiguous (several fit, and the LSTM ranked within that set) or unmatched (no canonical rule fits, so the label is an attribution the LSTM makes alone). The asymmetry is severe and bounds how the compositional result may be read: AllC and AllD carry a substantial deduced share, whereas the TFT and WSLS labels are almost entirely attributions.

Label	n	Deduced (%)	Ambiguous (%)	Unmatched (%)
AllC	11,312	23.6	36.9	39.6
TFT	2,277	5.4	0.0	94.6
WSLS	2,972	2.3	0.6	97.1
AllD	7,439	35.6	0.0	64.4

Table B.10. Validation of the LSTM branch. Trained on synthetic AllC / AllD / Tit-for-Tat / Win-Stay-Lose-Shift trajectories at zero and 5% execution noise, balanced at 40,320 sequences per strategy per level. The 10% and 20% noise levels are never trained on. No language-model transcript enters training.

Split	n	Accuracy
held-out (NoNoise + Noise005)	64,512	0.9846
OOD (Noise01)	161,280	0.9364
OOD (Noise02)	161,280	0.8107
held-out recall: AllC	16,150	0.9853
held-out recall: AllD	16,203	0.9952
held-out recall: TFT	16,193	0.9744
held-out recall: WSLS	15,966	0.9834

Table B.11. Pairwise correlation between the model curves, five models. Correlation of the ten-point cooperation-against-scale curves of each pair of the five models the main text reports. Six of the ten pairs are negative and the mean is -0.10, which is why averaging the curves cancels much of the movement.

	Claude Haiku	GPT-5.4 Nano	Gemini 3.5	Qwen3 235B-A22B	Grok 4.20
Claude Haiku 4.5		$+0.11$	-0.18	-0.56	$+0.40$
GPT-5.4 Nano	$+0.11$		-0.23	-0.72	$+0.85$
Gemini 3.5 Flash-Lite	-0.18	-0.23		$+0.22$	-0.15
Qwen3 235B-A22B	-0.56	-0.72	$+0.22$		-0.71
Grok 4.20	$+0.40$	$+0.85$	-0.15	-0.71	

Table B.12. Pairwise correlation between the model curves, all six models. The same correlations with the sixth model included: nine of the fifteen pairs are negative and the mean is -0.08.

	Claude Haiku	GPT-5.4 Nano	Gemini 3.5	Qwen3 235B-A22B	Grok 4.20	Gemini 3.1
Claude Haiku 4.5		$+0.11$	-0.18	-0.56	$+0.40$	-0.03
GPT-5.4 Nano	$+0.11$		-0.23	-0.72	$+0.85$	-0.70
Gemini 3.5 Flash-Lite	-0.18	-0.23		$+0.22$	-0.15	$+0.36$
Qwen3 235B-A22B	-0.56	-0.72	$+0.22$		-0.71	$+0.73$
Grok 4.20	$+0.40$	$+0.85$	-0.15	-0.71		-0.57
Gemini 3.1 Flash-Lite	-0.03	-0.70	$+0.36$	$+0.73$	-0.57	

Table B.13. Range of cooperation across the ten payoff scales, on the opening move and afterwards. At round 1 the agent has seen only the payoff matrix and the instructions, so a dependence on the payoff scale there cannot have come through the history of play. The two windows rest on 400 and 3,600 decisions per cell, so their ranges are not variance-matched and the ratio is descriptive rather than a test; the bias runs the same way in every model, while the ratio does not.

Model	Round 1	Rounds 2–10	Ratio
Claude Haiku 4.5	0.180	0.088	2.04
GPT-5.4 Nano	0.305	0.239	1.28
Gemini 3.5 Flash-Lite	0.367	0.194	1.90
Qwen3 235B-A22B	0.050	0.105	0.47
Grok 4.20	0.762	0.622	1.23
Gemini 3.1 Flash-Lite	0.252	0.069	3.68

C. The evolutionary baseline

The main text reports what six language models did. This appendix computes what the same payoff matrices imply for a population of adaptive agents restricted to the same four canonical rules, on the same ten payoff scales, and compares the two. The purpose is to size a gap rather than to test either object: the baseline says which strategy mixtures are evolutionarily stable when the only sources of variation are payoff magnitude and stochastic execution, and the difference between that and the corpus is a measurement of everything a language model brings that an evolutionary model does not. Combining evolutionary modelling with language-model simulation in this way has proved informative in recent work on governance dilemmas [4, 5].

C.1 The model

We adopt the standard pairwise-comparison process under the small-mutation limit [6–9], in which a finite well-mixed population of size $Z = 100$ evolves through social imitation governed by the Fermi update rule. The small-mutation limit is the regime in which mutations are rare enough that the population spends almost all of its time in monomorphic states, every individual playing the same strategy, and only occasionally transitions between them; it has a clean analytical solution and is the standard reference baseline in finite-population evolutionary game theory, and it has proved able to reproduce features of human experimental data [10, 11]. Selection intensity is set to $\beta = 0.1$.

To put the analysis on the same incentive structure as the experiments, the FAIRGAME penalty matrix is negated, so that lower-is-better penalties become higher-is-better utilities. The penalties $T_{\text{pen}} = 0$, $R_{\text{pen}} = 2$, $P_{\text{pen}} = 6$, $S_{\text{pen}} = 10$ map to utilities $T = 0$, $R = -2$, $P = -6$, $S = -10$, which satisfy the standard prisoner's dilemma ordering $T > R > P > S$; the transformation is a strict sign flip and preserves the strategic ordering. The whole matrix is then multiplied by λ, exactly as in the experimental protocol, so that the two sides of the comparison are anchored to the same per-round magnitudes.

For a resident strategy i and a mutant j, the expected per-round payoff π_{ij} is obtained analytically over the same horizon the experiments use, $r = 10$ rounds, by propagating the joint distribution over the two players' actions forward round by round and averaging; no simulation enters. The fixation probability of a single j-mutant in a resident i-population is

$$\rho_{ji} = \left(1 + \sum_{k=1}^{Z-1} \prod_{m=1}^{k} \frac{T_{ji}^{-}(m)}{T_{ji}^{+}(m)}\right)^{-1}, \tag{A 1}$$

where $T_{ji}^{\pm}(m)$ are the Fermi-weighted transition rates at composition m [12]. Assembling these probabilities gives an embedded Markov chain over the four monomorphic states whose unique stationary distribution σ is the long-run frequency of each strategy in the rare-mutation regime [6]. The reference implementation is the open-source EGTtools library [13].

C.2 Deterministic continuous dynamics and simplex orbits

In the infinite-population deterministic limit, evolutionary dynamics is described by the standard replicator equation [6, 14]:

$$\dot{x}_i = x_i \left[(\mathbf{A}\mathbf{x})_i - \mathbf{x}^T \mathbf{A}\mathbf{x}\right], \tag{A 2}$$

where $\mathbf{x} = (x_1, \ldots, x_4)^T$ is the population state on the three-simplex Δ_3 spanned by the four canonical rules (AllC, AllD, TFT, WSLS), and $\mathbf{A}$ is the 4×4 payoff matrix. Under a scalar payoff transformation $\mathbf{A}' = \lambda\mathbf{A}$ with $\lambda > 0$, the vector field scales uniformly:

$$\dot{x}_i' = \lambda\, \dot{x}_i. \tag{A 3}$$

Multiplying the payoff matrix by a positive constant multiplies the replicator vector field by that constant. This changes the velocity of state evolution along trajectories by a uniform time reparametrisation $d\tau = \lambda\, dt$, but leaves every trajectory, orbit geometry, invariant manifold, and rest point strictly invariant.

Figure C.1 illustrates this mathematical invariance across the four faces of the three-simplex spanned by the canonical rules, leaving one rule out per face. Across 2,000 randomly sampled population states, the normalised vector fields at $\lambda = 0.01$ and $\lambda = 1000$ agree to within 4×10^{-14}, confirming that continuous deterministic selection cannot account for any empirical dependence on payoff scale.

C.3 The strategies under execution noise

The four canonical rules enter as memory-one policies with a per-round implementation error ε, the probability of executing the opposite of the intended action. Writing $\text{hi} = 1 - \varepsilon$ and $\text{lo} = \varepsilon$, the probability of cooperating is hi wherever

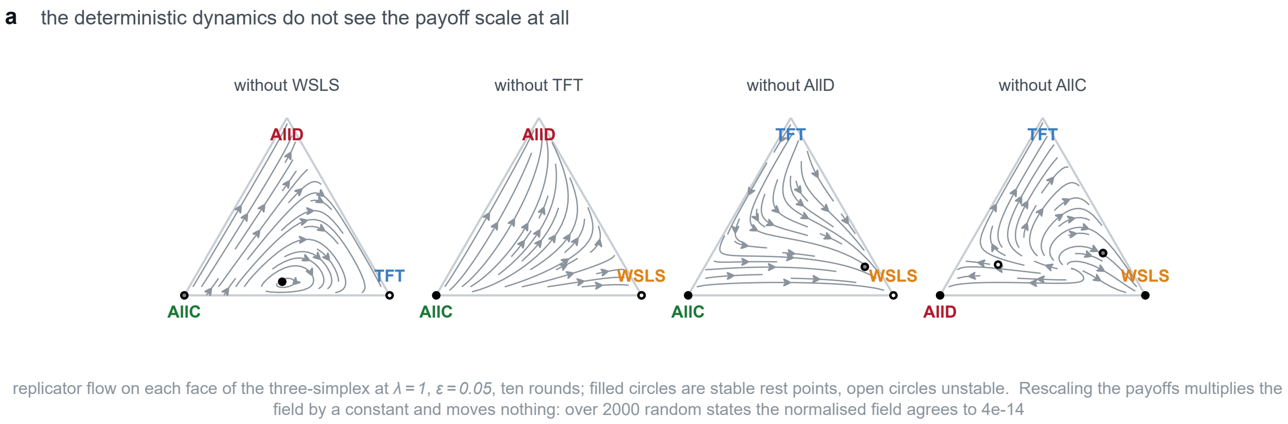

Figure C.1. The deterministic dynamics do not see the payoff scale. Replicator flow on each of the four faces of the three-simplex spanned by the canonical rules, one face per strategy left out, at $\lambda = 1$, $\varepsilon = 0.05$ and ten rounds. Filled circles are stable rest points, open circles unstable. Only one row is drawn because rescaling the payoffs multiplies the field by a constant and moves nothing: the matrix at $\lambda = 1000$ is exactly 10^5 times the matrix at $\lambda = 0.01$, and over 2,000 random population states the normalised fields agree to 4×10^{-14}.

table A.3 prescribes C and lo wherever it prescribes D, including on the opening move, so AllC, Tit-for-Tat and Win-Stay-Lose-Shift open at hi and AllD at lo. The rules are therefore the same objects the empirical read-out matches against, which is what makes the two sides of the comparison commensurable.

C.4 Numerical method and precision audit

The computation is numerically delicate at the top of the payoff grid, and we report the audit rather than only the result. Equation (A 1) accumulates a product of $Z - 1$ Fermi ratios whose exponents scale with $\beta\lambda\Delta\pi$; at the largest scales that product overflows the range of double precision, every fixation probability rounds to exactly 0 or 1, the embedded chain acquires more than one absorbing state, and the eigenvector for eigenvalue 1 is no longer unique. A double-precision solver then returns an arbitrary member of a degenerate eigenspace, which is a wrong answer that carries no warning.

The published grid is therefore computed twice: once in arbitrary precision with `mpmath` at 200 decimal digits, which is the primary solver, and once in double precision through EGTtools, which serves as a check. Each cell of the grid records the maximum absolute discrepancy between the two solvers and the number of absorbing states the double-precision chain possesses, so a reader can see exactly where double precision fails and by how much.

C.5 The grid

Figure C.4 draws the stationary frequency of each canonical strategy against the payoff scale at every execution-noise level, and the tables below give the same grid in numbers, together with the precision audit.

C.6 Finite-population invasion structure across scales

In contrast to the continuous deterministic limit, the payoff scale λ acts directly on finite populations governed by stochastic imitation. Under the Fermi update rule, the probability that an individual adopting strategy i switches to strategy j upon pairwise comparison is:

$$P(i \to j) = \frac{1}{1 + \exp\left(-\beta(\pi_j - \pi_i)\right)} = \frac{1}{1 + \exp\left(-\beta\lambda(\pi_j^{(0)} - \pi_i^{(0)})\right)}, \tag{A 4}$$

where $\pi_j^{(0)}$ denotes baseline unscaled payoffs. The payoff scale λ and the selection intensity β enter the update rule exclusively through their product $\beta\lambda$. Consequently, scaling the payoffs by λ is formally equivalent to scaling the selection pressure.

Figure C.2 shows the resulting invasion graphs across the four monomorphic states as λ varies. At $\lambda = 0.01$, selection is negligible, and transition probabilities approach neutral drift ($1/Z = 0.01$). As λ increases to baseline ($\lambda = 1.0$) and high stakes ($\lambda = 10.0$), selection sharpens, making AllD an evolutionary sink that resists invasion from all three cooperative rules and fixes with high probability.

Table C.1. Stationary distribution of the evolutionary baseline at $\varepsilon = 0.05$. Long-run frequency of each canonical rule under the pairwise-comparison process in the small-mutation limit, $Z = 100$, $\beta = 0.1$, $r = 10$ rounds, computed in 200-digit arithmetic. The last two columns are the precision audit: the largest absolute disagreement between the 200-digit solver and a double-precision one, and the number of absorbing states the double-precision chain acquires where that number exceeds one. Double precision fails exactly where the audit says it does, at the top of the grid, and returns a wrong answer there without any warning.

	stationary frequency				audit	
λ	AllC	AllD	TFT	WSLS	max $\lvert\Delta\rvert$	absorbing
0.01	0.228	0.267	0.251	0.253	4.8e-15	0
0.1	0.100	0.382	0.260	0.258	6.0e-15	0
0.25	0.049	0.446	0.258	0.247	3.6e-15	0
0.5	0.032	0.507	0.224	0.237	1.5e-14	0
1	0.022	0.606	0.150	0.223	1.7e-12	0
2	0.013	0.738	0.068	0.181	1.9e-07	0
5	0.002	0.894	0.010	0.094	4.1e-08	0
10	2×10^{-4}	0.969	0.001	0.030	1.1e-12	0
100	4×10^{-26}	1.000	7×10^{-26}	4×10^{-12}	1.0e+00	2
1000	2×10^{-245}	1.000	4×10^{-245}	4×10^{-112}	1.0e+00	2

Table C.2. Stationary frequency of Always Defect, across the payoff grid and every execution-noise level. The same process as table C.1. Raising the payoff scale hands the population to Always Defect at every noise level, and does so sooner the noisier execution is. The one exception to monotonicity is the error-free corner $\varepsilon = 0$, where the share dips slightly before rising: with no execution error Tit-for-Tat loses only its opening move to Always Defect, so raising the scale first sharpens selection in Tit-for-Tat's favour. Any positive error rate removes the exception, because it lets Always Defect exploit the retaliation lag repeatedly rather than once.

	execution error ε				
λ	0	0.05	0.1	0.2	0.3
0.01	0.263	0.267	0.269	0.268	0.263
0.1	0.319	0.382	0.416	0.429	0.394
0.25	0.315	0.446	0.546	0.634	0.602
0.5	0.304	0.507	0.661	0.811	0.822
1	0.283	0.606	0.802	0.939	0.965
2	0.273	0.738	0.937	0.993	0.999
5	0.347	0.894	0.998	1.000	1.000
10	0.563	0.969	1.000	1.000	1.000
100	1.000	1.000	1.000	1.000	1.000
1000	1.000	1.000	1.000	1.000	1.000

Table C.3. The evolutionary baseline beside the corpus, at matched payoff scales. Left, the stationary frequency of each canonical rule at $\varepsilon = 0.05$; right, the share of agent-games carrying each label, pooled over the five models the main text reports. The two move in opposite directions: the baseline's Always Defect share rises from 0.267 to 1.000 across the grid while the corpus's falls from 0.436 to 0.283.

	evolutionary baseline				corpus			
λ	AllC	AllD	TFT	WSLS	AllC	AllD	TFT	WSLS
0.01	0.228	0.267	0.251	0.253	0.381	0.436	0.091	0.092
0.1	0.100	0.382	0.260	0.258	0.398	0.395	0.094	0.113
0.25	0.049	0.446	0.258	0.247	0.505	0.266	0.107	0.122
0.5	0.032	0.507	0.224	0.237	0.434	0.303	0.110	0.152
1	0.022	0.606	0.150	0.223	0.472	0.277	0.099	0.152
2	0.013	0.738	0.068	0.181	0.501	0.265	0.105	0.130
5	0.002	0.894	0.010	0.094	0.471	0.272	0.114	0.142
10	2×10^{-4}	0.969	0.001	0.030	0.496	0.271	0.108	0.124
100	4×10^{-26}	1.000	7×10^{-26}	4×10^{-12}	0.412	0.307	0.114	0.166
1000	2×10^{-245}	1.000	4×10^{-245}	4×10^{-112}	0.458	0.283	0.127	0.133

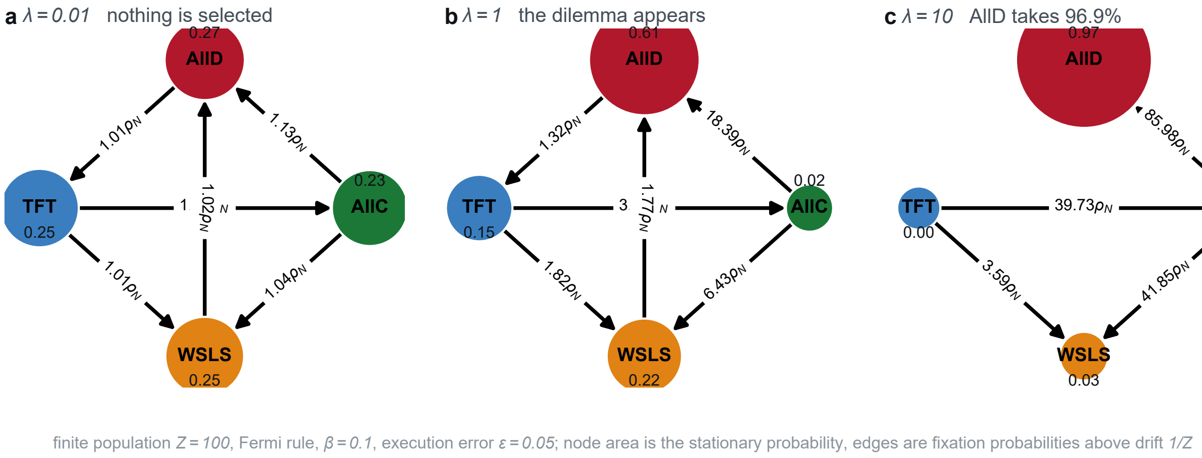

Figure C.2. Where the payoff scale does act: the invasion structure of a finite population. Nodes are the four monomorphic states, node area is the small-mutation-limit stationary probability, and an edge is drawn where a single mutant fixes more often than neutral drift $1/Z$ would carry it, labelled in multiples of that drift. $Z = 100$, Fermi pairwise comparison, $\beta = 0.1$, $\varepsilon = 0.05$. Because λ and β enter the update rule only through their product, raising the scale is raising selection pressure. The panels stop at $\lambda = 10$ because the float64 fixation probabilities underflow beyond it; those cells are solved at 200 digits instead.

C.7 Comparison with the corpus

Figure C.3 places the stationary distribution beside the pooled label composition of the corpus at each payoff scale.

Two features agree. Both put their greatest behavioural diversity at the smallest payoff scales, where the baseline is close to neutral drift because $\beta\lambda\Delta\pi$ is small and the corpus shows its models at their most dissimilar. And in both, a change in λ moves mass between the conditional cooperators and unconditional defection rather than reshuffling the four strategies arbitrarily.

The direction is the disagreement, and it is the whole point of the comparison. The baseline drives Always Defect upward as the payoff scale grows, because a larger λ multiplies the payoff differences that the Fermi rule responds to and so sharpens selection against any rule that can be exploited; at $\varepsilon = 0.05$ its share runs from 0.267 at $\lambda = 0.01$ to 1.000 by $\lambda = 100$, and the process ends with the population entirely on Always Defect (table C.3). The corpus does the opposite in aggregate: its pooled AllD share falls from 43.6% at $\lambda = 0.01$ to 28.3% at $\lambda = 10^3$, and its AllC share rises from 38.1% to 45.8%. The two objects, given the same matrices and restricted to the same four rules, predict qualitatively different worlds depending on who is playing.

One qualification belongs with the baseline rather than with the comparison. The rise of Always Defect in λ is monotone at every positive execution error, but not in the error-free corner $\varepsilon = 0$, where the share falls from 0.319 at $\lambda = 0.1$ to 0.273 at $\lambda = 2$, a dip of 0.046, before rising (table C.2). The reason is specific and worth stating, because it is a property of the model and not of the arithmetic. With no execution error, AllC, Tit-for-Tat and Win-Stay-Lose-Shift earn exactly -2λ against each other and against themselves, so the three cooperative rules are neutral among themselves and the ordering is decided entirely by their exchanges with Always Defect. Over ten rounds Tit-for-Tat loses only its opening move to Always Defect, whereas Win-Stay-Lose-Shift keeps being re-exploited and AllC is exploited throughout, so raising λ first sharpens selection in Tit-for-Tat's favour before Always Defect's one-round advantage finally decides it. Any positive error rate removes the dip, because it lets Always Defect exploit the retaliation lag repeatedly rather than once. The empirically implied error rates are all positive and well above zero (§C.8), so the qualification does not touch the comparison.

A second disagreement is structural rather than directional. The baseline is language-blind by construction: there is nothing in equation (A 1) for a prompt language to enter. In the corpus, the interaction of the model with the prompt language carries 17.0% of the variance across cells, more than the payoff scale carries as a main effect. Whatever the corpus is responding to, a payoff matrix is not a complete description of it.

We read the gap as a measurement and not as a falsification. It sizes what these agents bring that an evolutionary model does not, from alignment training to the cooperative bias of a human text corpus and the connotations the prompt language carries. Treating the baseline as the null and the gap as the quantity to be explained is, we would suggest, a productive way to read studies of language models in strategic settings.

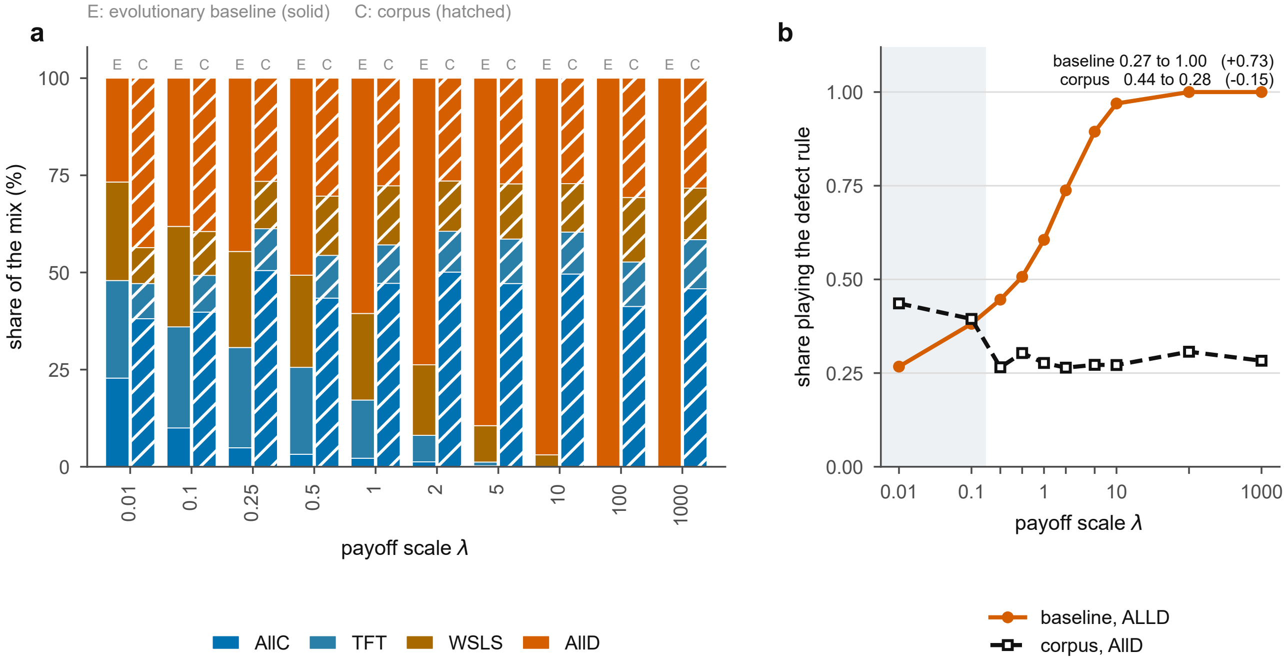

Figure C.3. The evolutionary baseline against the corpus, at matched payoff scales. (a) Stationary frequency of each canonical strategy under the small-mutation limit ($Z = 100$, $\beta = 0.1$, $r = 10$, $\varepsilon = 0.05$), beside the pooled label composition of the five main-text models at the same payoff scale. (b) The share playing Always Defect alone, the one series on which the two disagree in sign: across the ten scales the baseline rises from 0.27 to 1.00 while the corpus falls from 0.44 to 0.28.

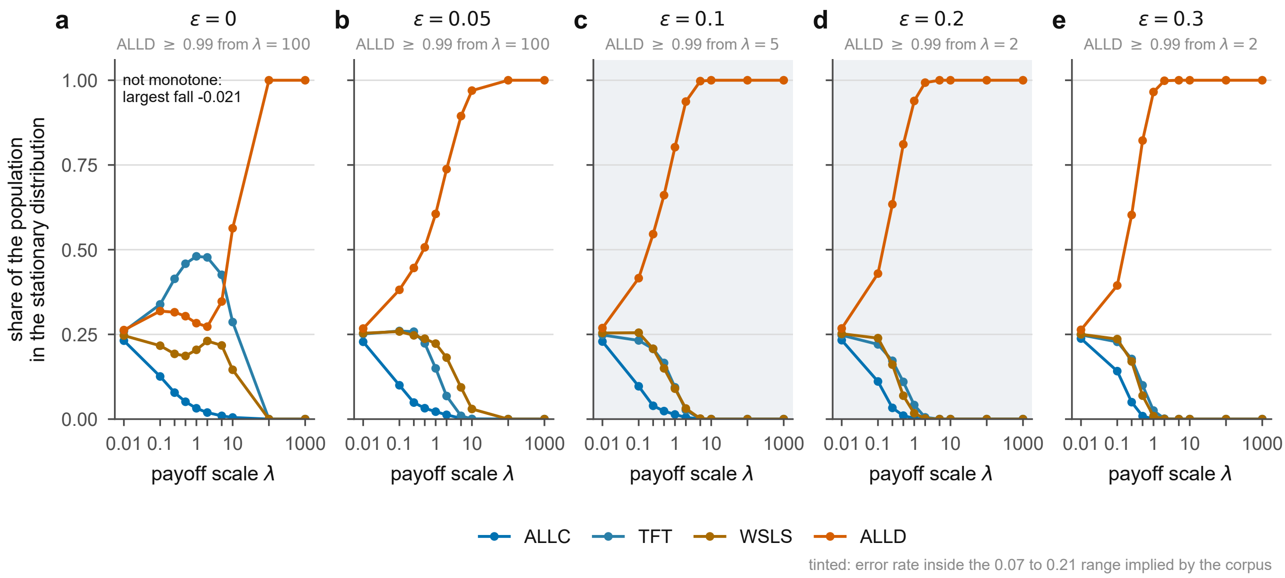

Figure C.4. The evolutionary baseline across the payoff grid and the execution-noise levels. Stationary frequency of each of the four canonical strategies against the payoff scale, one panel per execution-error rate ε. Always Defect rises with λ at every noise level, and does so sooner the noisier execution is; the rise is monotone at every positive ε and dips slightly before rising in the error-free corner $\varepsilon = 0$ (table C.2).

C.8 Execution noise is not a free parameter here

In a study with no empirical read-out, ε would have to be chosen. Here it can be estimated, because the rule base already measures how far each model's play sits from the nearest canonical rule. Dividing a model's mean rule distance by the ten rounds of a game gives a per-round rate at which its realised play departs from the rule it most resembles, which is the empirical analogue of an implementation error (table C.4).

Table C.4. Execution noise implied by the read-out. The mean number of rounds, out of ten, by which an agent-game departs from the nearest canonical rule, and that quantity as a per-round rate. The two models whose play is least rule-governed imply a noise level at the very edge of the range the classifier was validated on, which is why §D.3 treats their strategy labels as the weakest evidence in this article.

Model	Mean rule distance (rounds of 10)	Implied ε
Claude Haiku 4.5	2.073	0.207
GPT-5.4 Nano	2.120	0.212
Gemini 3.5 Flash-Lite	1.077	0.108
Qwen3 235B-A22B	0.737	0.074
Grok 4.20	0.725	0.073
Gemini 3.1 Flash-Lite	1.030	0.103

The implied rates span 0.073 to 0.212, so no single noise level matches the whole corpus, and the honest matched comparison for two of the five main-text models is at $\varepsilon = 0.2$ rather than at the 0.05 a study of this kind would ordinarily adopt. The grid in §C.5 therefore carries that level, and the conclusion of §C.7 is unchanged across it: the direction of the baseline in λ does not depend on ε.

This mapping should not be over-read. A symmetric per-action flip probability is a clean theoretical object and a coarse abstraction of what makes a language model's play depart from a rule, which includes prompt sensitivity, end-game reasoning and the heterogeneity documented throughout this article. The calibration bounds the comparison; it does not fit the model.

D. Robustness

D.1 What including the sixth model changes

Every pooled quantity in this article is reported twice, once on the five main-text models and once on all six, so that the effect of the split can be seen rather than assumed. Including Gemini 3.1 Flash-Lite moves the pooled range from 0.154 to 0.125, the mean within-model range from 0.249 to 0.219 and the attenuation from 1.62 to 1.76; the share of anticorrelated pairs rises from six of ten to nine of fifteen; and in the variance decomposition the payoff scale falls from 5.8% to 4.3% as a main effect while its interaction with the model rises from 17.9% to 19.1% (tables B.4, B.5, B.11 and B.12).

Every sign, every significance and every ordering agrees between the two, and the pooling argument the main text makes is stronger over all six models than over the five it reports. The single count that the sixth model changes is the strategy-trend tally, and it changes it in the direction that weakens the paper, which is stated in the main text at the point where the claim is made rather than only here.

D.2 Robustness of the evolutionary baseline across 80 settings

The finding that the evolutionary baseline predicts an increase in defection with payoff scale is not an artefact of the specific baseline parameters ($Z = 100$, $\beta = 0.1$, $\varepsilon = 0.05$). To confirm this, we solved the stationary distribution across a comprehensive grid of 80 parameter combinations: four population sizes $Z \in \{50, 100, 200, 500\}$, five selection intensities $\beta \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$, and four execution noise levels $\varepsilon \in \{0, 0.05, 0.1, 0.2\}$, across all ten payoff scales (800 cells in total, each computed at 200 decimal digits).

Figure D.1 presents this parameter sweep. Across all 80 settings, the stationary share of Always Defect increases with λ, exceeding 90% at large scales in every setting, with a median increase of 0.716 and a minimum increase of 0.195. In 61 of the 80 settings the rise is strictly monotone; the 19 non-monotone cases occur exclusively at $\varepsilon = 0$, where a transient dip occurs before selection drives AllD to fixation (§C.7). Crucially, the empirical LLM corpus moves in the opposite direction to this theoretical prediction across all evaluated models.

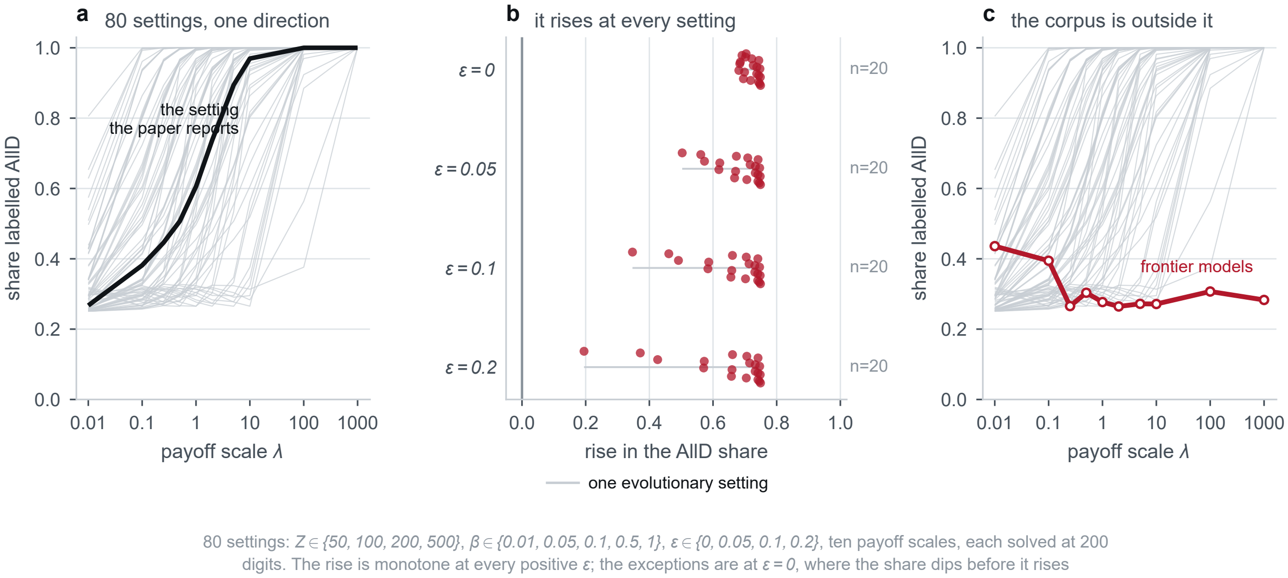

Figure D.1. The baseline's direction is a property of the model, not of its parameters. (a) The AllD share against the payoff scale, one hairline per evolutionary setting, with the setting reported in the text drawn heavy. (b) The rise in the AllD share from the smallest scale to the largest, by execution error. (c) The corpus on the same axis as the family. Eighty settings, $Z \in \{50, 100, 200, 500\}$ crossed with $\beta \in \{0.01, 0.05, 0.1, 0.5, 1\}$ and $\varepsilon \in \{0, 0.05, 0.1, 0.2\}$, ten payoff scales, 800 cells each solved at 200 significant digits. The rise is monotone at every positive ε; all 19 exceptions are at $\varepsilon = 0$.

D.3 The read-out at the edge of its validated range

The classifier was validated to 20% execution noise, which is two rounds in ten. Table C.4 shows that Claude Haiku 4.5 and GPT-5.4 Nano sit at 2.07 and 2.12 rounds from the nearest canonical rule on average, so for those two models the read-out is operating at the very edge of the range it was validated on, and their label compositions are the weakest evidence in this article.

The two claims that matter most are insulated from this. The falling share of agent-games matched exactly by one rule, and the rising number of rounds violating the nearest rule, are both computed by exact matching with no learned component at all, and it is precisely those two quantities that the main text uses to argue that play becomes less rule-governed. The compositional claim, which does use the classifier, is reported with its provenance beside it (table B.9) and is described in the main text as the weaker of the two.

There is a further and more general caveat. A strategy is a decision rule and not a decision, so it is never observed; separating the memory-one rules from observed play has been shown to require a long run of observations even where the resident rule is fixed by construction [15, 16]. Our trajectories are ten rounds long. This is why the read-out is reported throughout as a resemblance and never as an identification, and why richer strategy vocabularies, including zero-determinant and extortionate play [17], and mixture estimators designed for longer horizons [18, 19], are named as the natural extension rather than attempted here.

D.4 Parsing, fallback and the direction of the bias

The parsing fallback records an unresolvable reply as Option A, which is defection under this framing (§A.2). Any contamination it introduces therefore lowers a measured cooperation rate. Two consequences follow. The cooperative behaviour reported in the main text cannot be an artefact of the fallback, because the fallback cannot produce cooperation. And if the fallback rate were correlated with the payoff scale, it would attenuate a measured range rather than create one, so the smallest ranges in table 1 are the ones such a correlation could most plausibly distort, and they are also the ones the article leans on least.

D.5 What this design does not measure

Two absences are worth naming so that they are not mistaken for findings. First, there is no measured test-retest floor: our reference for calling a departure large is the sampling variability of these data, through a bootstrap over dyads and a permutation null, and not an independent repeat of the whole protocol under identical conditions. Such a repeat is the obvious next measurement, and until it exists the ranges in table 1 should be read against the permutation null rather than against an unmeasured reproducibility baseline.

Second, clustering on the dyad handles the dependence between the two agents of a game and between the rounds inside an agent-game, but it is not a mixed-effects model with random intercepts for replicate and game, which would be the fuller treatment of a nested design. Because the direction of every effect is consistent across conditions and the clustered intervals are wider than their unclustered counterparts, we regard the reported inference as conservative rather than optimistic, but we do not claim it is complete.

E. Reproduction

Every number, figure and table in this article and in this appendix is produced by an archived pipeline from the released transcript corpus, and no value is retyped by hand into the manuscript sources: the three main-text tables are generated into `tables_auto.tex`, the supplementary tables of §B into `supp_tables_auto.tex`, and the evolutionary grid of §C into `egt_tables_auto.tex`.

The pipeline runs in the following order. `s00_build` reads the transcript corpus, applies the completeness guard of §A.1 and writes the tidy game, round and read-out tables. `s01_train_lstm` trains the classifier of §A.6 on the synthetic corpus. `s02_readout` applies the rule base and the classifier to the transcripts. `s03_stats` and `s04_strategy_stats` compute every statistic reported here. `s05_figures` and `s05b_appendix_figures` draw the figures directly into the manuscript directory. `s06_tables` and `s07_supplementary` render the tables. `s09_egt` computes the evolutionary grid of §C. Finally `s08_verify_paper` recomputes every quantity quoted in the manuscript prose from the data and fails if any of them disagrees, so that a stale number cannot survive an edit.

Three seeds fix the stochastic parts: 12345 for the corpus build, 1234 for classifier training, and 20260905 for the bootstrap and permutation draws.

The results reported here were produced under Python 3.12.3 with NumPy 1.26.4, pandas 2.2.2, PyArrow 16.1.0, SciPy 1.17.1, statsmodels 0.14.6, scikit-learn 1.5.1, PyTorch 2.11.0 on CPU, Matplotlib 3.9.2, mpmath 1.3.0 and EGTtools 0.1.14.2. The classifier is trained on CPU and its seed is fixed, so the read-out is reproducible without a GPU.

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